Bonded knots

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Abstract

The main objective of this paper is to present a systematic classification of uncolored bonded knots with singularity number at most nine. Bonded knots can model proteins that contain strong bonds between distant amino acids, such as ionic bonds or disulfide bridges. We describe the methods used to achieve this classification, including the generation of bonded knot diagrams and the distinction of topological knotted types through Yamada polynomial and Reidemeister moves.

1 Introduction

Proteins perform many functions essential to life processes, including signal transduction, gene transcription, and catalysis of chemical reactions. Among these, catalytic activity is particularly prominent when it comes to knotted proteins - rare biomolecules whose backbones entangle themselves in a knot. Notably, roughly 80% of knotted proteins are enzymes [Source], while others have been identified as membrane proteins and secondary active transporters[source]. Despite ongoing research, their functions and folding mechanisms are still not well understood.

Since protein function is directly related to its conformation, an analysis of their structure may prove helpful in answering these questions. Bonded knots can be applied to model such structures. This approach captures not only the spatial knotting of the backbone but also incorporates the chemical intra-chain bonds, representing disulfide bridges, hydrogen bonds, and salt bridges. These additional constraints are represented by embedded arcs, which distinguish bonded knots from classical knots. More generally, collections of multiple chains connected by additional intra- or intermolecular bonds can be modeled as bonded links. Since protein backbones are not closed curves, a suitable closure must be introduced to enable the application of knot-theoretic methods.

Notably, this approach extends beyond knotted proteins and applies to a broader class of biomolecules containing intra-chain or intermolecular bonds. For instance, proteins with disulfide bonds alone constitute roughly 30% of all proteins [Source], highlighting the potential relevance of bonded knots in structural biology.

As of now, bonded structures have been classified up to 7 singularities (number of crossings and vertices) [source]. In this paper, we extend this classification to 9 singularities and describe the methodology used to obtain these results.

2 Defintions

[source: https://arxiv.org/pdf/2603.18768;]

A *graph* is a collection of edges and vertices such that each edge has a pair of vertices as its endpoints. A *spatial graph* is a graph embedded in 3-space.

(pisac o 3-valent??)

[https://arxiv.org/pdf/1910.04398 ¡- dla colored, ale skopiowalam]

A *bonded knot* is a tuple (L, b), where

- L is a knot embedded in the three-sphere S^3 (or R^3),
- $b = \{b_1, b_2, \ldots, b_n\}$ is a collection of pairwise disjoint bonds (closed intervals) properly embedded in $S^3 \setminus L$.

Bonded knots can be represented as 3-valent graphs equipped with a perfect matching.

A *bonded link* is defined analogously, where L is a link. The bonds may connect different components of the link or the same component.

2.1 Projections, Diagram Equivalence and Reidemeister Moves

[source: https://arxiv.org/pdf/1912.09353]

A *knot diagram* is a regular projection of a knot onto a plane, with crossings equipped with over–under information.

The *Planar Diagram (PD) code* of a bonded knot is a combinatorial encoding of a diagram obtained by labeling its arcs and recording, for each node (crossing or bond endpoint), the incident arcs in counterclockwise order. In the case of a crossing, the ordering is initiated at an undercrossing arc. The resulting PD code is an ordered list of such node entries that completely determines the planar diagram.

Two knot diagrams are *equivalent* if there exists a finite sequence of Reidemeister moves transforming one diagram into the other.

[wiecej o tym V(?) https://arxiv.org/pdf/2603.18768]

Reidemeister moves are local transformations of knot diagrams that preserve knot type. There are three classical Reidemeister moves (I–III), together with additional moves IV and V for bonded knots (or knotted graphs), which describe interactions at vertices where bonds meet the knot.

The *singularity number* is defined as

$$s(K) = c(K) + 2b(K),$$

where $c(K)$ denotes the minimal number of crossings among all diagrams of a bonded knot K under generalized Reidemeister moves, and $b(K)$ is the number of bonds in K.

2.2 Diagram Terminology

A *vertex* is a bond endpoint. A vertex is represented in PD-code as $V[a_1, a_2, a_3]$, where the entries denote the incident arcs listed in counterclockwise order.

A *crossing* is a point where one strand passes over another. In the PD-code representation, the incident arcs are ordered counterclockwise, starting at an undercrossing arc; undercrossing arcs occupy odd positions and overcrossing arcs even positions. A crossing is denoted by $X[a_1, a_2, a_3, a_4]$. .[https://arxiv.org/pdf/1912.09353]

A *node* is either a vertex or a crossing.

An *arc* is a smooth curve joining a pair of adjacent nodes, with no vertices or crossings in its interior.

A *strand* is a simple path in the diagram whose endpoints are nodes, and whose interior contains no vertices. Crossings may occur along a strand.

An *edge* is a strand between two vertices.

A *diagram reduction* occurs when a diagram with n nodes can be transformed, via Reidemeister moves, into an equivalent diagram containing strictly fewer than n nodes.

3 Classification method

How Am I to Know

The classification method used in this thesis largely follows the scheme presented in [...] It involves generating all possible diagrams with vertices of degree 3 and 4, converting 4-valent vertices into crossings of both types (over- and under-crossings), computing invariants using the Yamada polynomial, and finally simplifying diagrams belonging to the same Yamada equivalence group in order to obtain unique diagrams.

The simplification algorithm presented in [...] uses sequences of Reidemeister moves to explore sets of diagrams reachable from the input states and performs a brute-force comparison, identifying two diagrams as equivalent whenever these sets intersect.

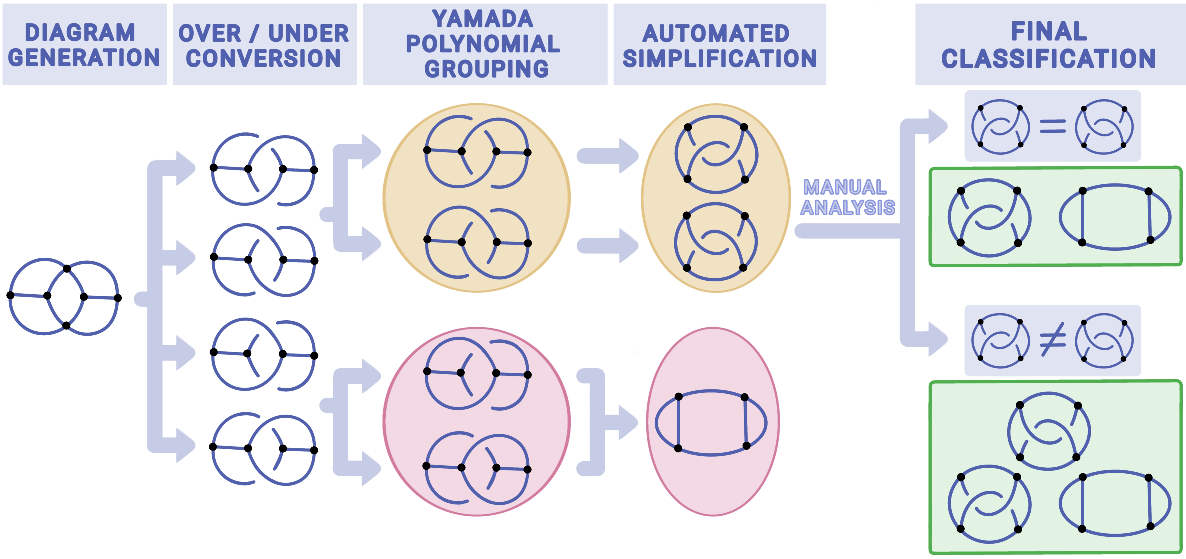


Figure 1: Overview of the proposed methodology and classification workflow.

The search is limited by a prescribed crossing-increasing depth in the Reidemeister move space. Increasing this depth allows more equivalent diagrams to be identified, but also makes the search more computationally expensive.

The number of diagrams to be considered grows rapidly with the number of singularities. Therefore, in order to classify bonded knots with more than seven singularities, we modified the simplification approach to improve its efficiency.

4 Simplification Idea

Come Together

Instead of applying many different sequences of Reidemeister moves, we attempt to transform diagrams toward their outerplanar forms. This approach should allow equivalent diagrams to be identified more efficiently, since each move brings the diagrams closer to a common canonical form, rather than exhaustively exploring the Reidemeister move space.

Naturally, a diagram may have multiple outerplanar forms, particularly because our method allows the number of crossings to increase (see Section 2). Therefore, it is still necessary to explore the Reidemeister move space; however, when the diagrams are equivalent, the explored space is much smaller than in the original approach.

Simplification Algorithm Overview

The input diagrams whose equivalence is to be determined are initially added to a queue. During each iteration, one diagram is removed from the queue and processed as follows:

- Its canonical form is computed and stored together with the input diagram from which it was obtained.
- If the same canonical form has previously been obtained from a different input diagram, the corresponding diagrams are identified as equivalent.
- Equivalence is treated as a transitive relation. Therefore, if diagrams A and B are equivalent, and diagrams C and D are equivalent, then discovering that B and C are equivalent merges the two equivalence classes into a single class containing A, B, C, and D.
- Diagrams already identified as outerplanar are discarded from further exploration.

- To prevent cycles, each visited state is stored as a pair consisting of its canonical form and the length of its outer face, since the same canonical form may occur with different outer-face configurations.
- Diagrams that have already been visited are not explored again. - Successor states are generated by applying transformations intended to guide the diagram toward an outerplanar form. During this process, all possible outer faces of the recorded length are considered.
- The generated diagrams are added to the queue.

The algorithm terminates when all diagrams belong to a single equivalence class, the search space is exhausted, or a predefined time limit is reached.

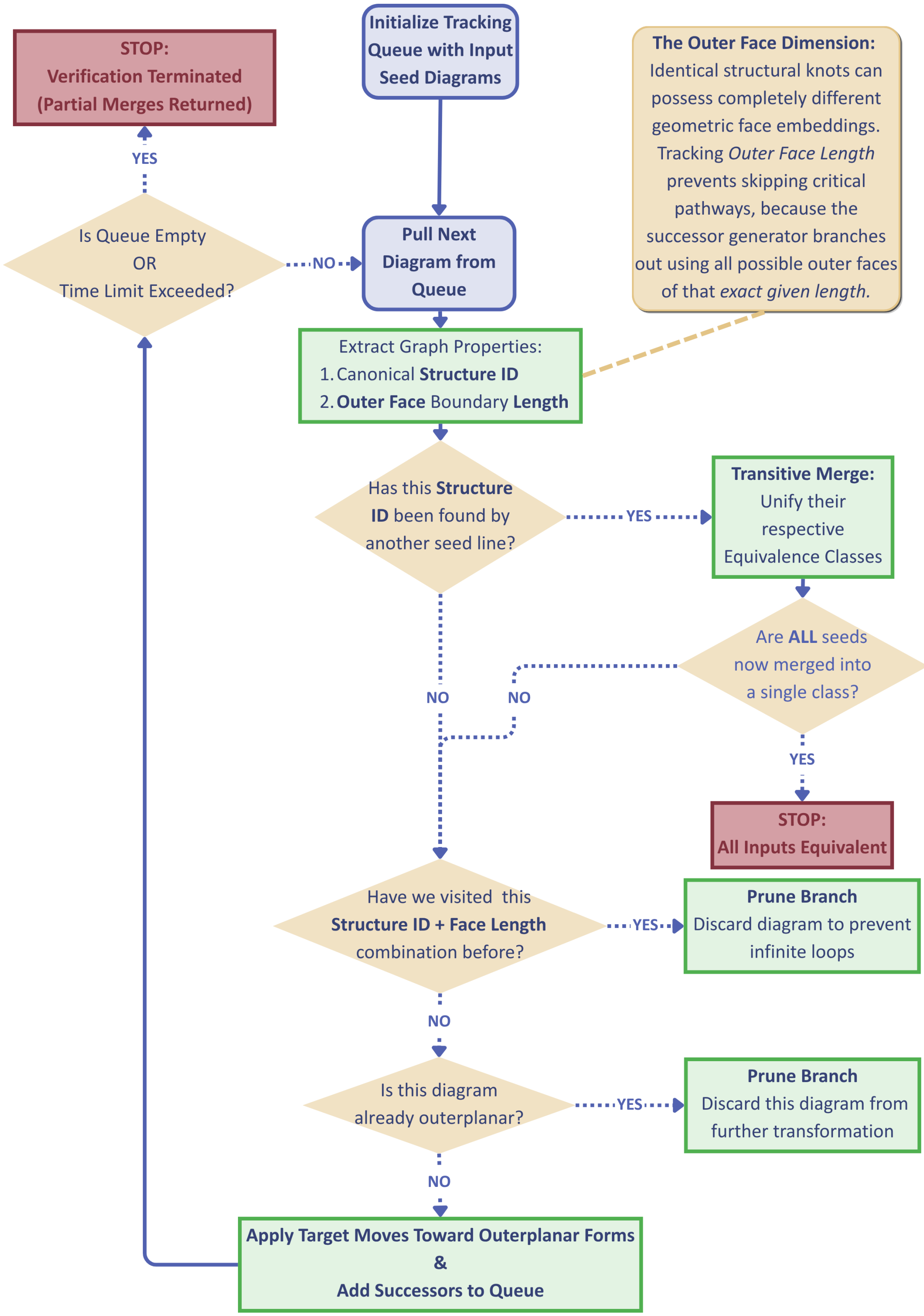


Figure 2: The visual workflow of the simplification algorithm.

4.1 Successor State Generation

I Want to Break Free (to the outer face)

The transformation process operates on a diagram with a selected outer face. Once the outer face has been fixed, the vertices lying in the interior of the diagram can be identified. The procedure then repeatedly performs the following steps:

1. Identify the inner vertex closest to the outer face.
2. Move this vertex toward the outer face through the nearest outer-face segment.
3. Reduce diagram: the transformation from step 2 often produces a diagram that can be simplified using the reduction procedure described in Section 3. The resulting diagram is therefore simplified repeatedly until no further reduction moves are applicable. Reduction moves that would return a newly introduced outer-face vertex to the interior of the diagram are not performed.

Although the transformation step aims to move additional vertices onto the outer face, subsequent reduction operations may alter the structure of the diagram in such a way that some vertices previously lying on the outer face become interior vertices again.

Cycles may occur in the search process, which is why previously visited states are stored, as described in Section 2.

The proposed procedure does not guarantee that an outerplanar form will always be reached. Nevertheless, equivalent diagrams may still be identified because the search process can lead to common intermediate states even when an outerplanar representation is not obtained.

[Obrazek]

4.1.1 Identifying the inner vertex closest to the outer face.

The interior vertices and the nodes lying on the outer face are first identified. These are then used as starting and ending points for the minimal-crossing path-finding function. The minimal-crossing path function (see Section ...) determines a path between a given pair of nodes that minimizes the number of crossings introduced by inserting a strand along the path.

For each target outer-face segment (i.e., the strand between two adjacent vertices), one path is determined. This procedure identifies not only the interior vertex closest to each outer-face segment, but also the corresponding paths along which the vertices can be moved onto the outer face.

[Obrazek]

4.1.2 Moving the Inner Vertex onto the Outer Face

Multiple paths are identified in the previous step, and a new diagram is generated for each of them as follows.

Since each vertex has three incident edges, this procedure requires inserting three new strands along the path associated with the selected interior vertex (see previous section). These strands are then connected outside the diagram to form a new outer-face vertex. Finally, the remaining ends of the inserted strands are connected to the corresponding endpoints of the original vertex, thereby integrating the new structure into the diagram.

Moving a vertex onto the outer face introduces new crossings. Whether these are assigned as overcrossings or undercrossings is determined by the immediate neighborhood of the vertex:

- If the majority of the three incident strands terminate at overcrossings, the newly introduced crossings are assigned as overcrossings.
- Otherwise, the newly introduced crossings are assigned as undercrossings.

This heuristic increases the likelihood that the reduction step (see Section ...) will remove some of the newly introduced crossings, thereby reducing the number of crossings retained in the final diagram.

[Obrazek]

4.1.3 Diagram-Reducing Strand Flip Move

Idea

When a strand crosses exclusively over all other strands it encounters in the diagram (Fig. 1), it is possible to perform a *strand flip* - move the strand to an alternative position in the diagram in such a way that all of its current crossings disappear. This operation may introduce new crossings over other strands, depending on the chosen placement (Fig. 2a). Consequently, if the alternative position induces fewer crossings than the original, then the total number of crossings in a diagram decreases and the diagram is therefore reduced (Fig. 2b, 2c).

The idea above is described for strands that pass exclusively over other strands they encounter. By symmetry, the same reasoning holds for strands that pass exclusively under all encountered crossings.

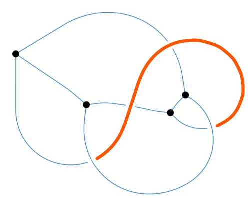
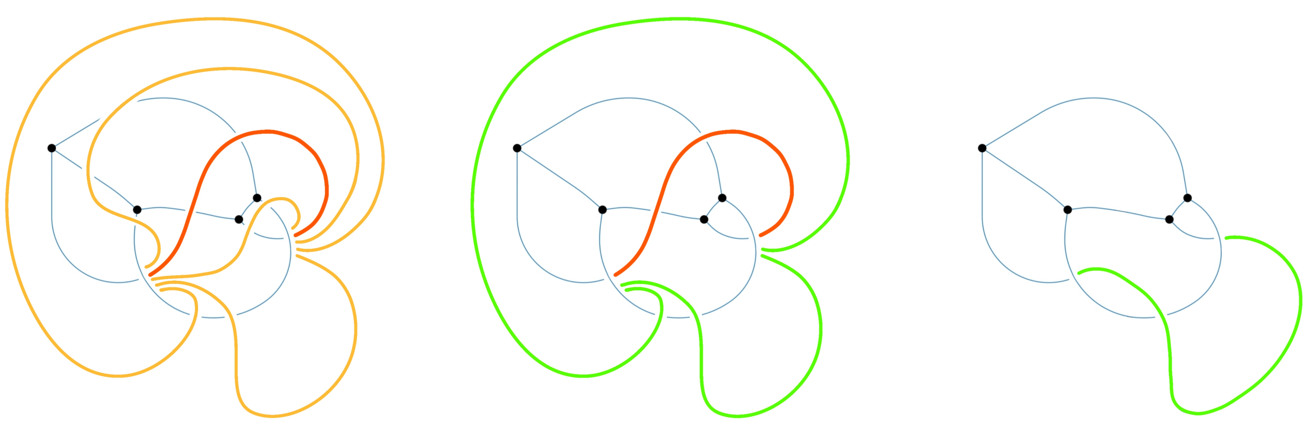


Figure 3: over-over path example



(a) Alternative path placement example

(b) better paths example

(c) Diagram after reduction

Figure 4: Caption

Algorithm: Detection of Diagram-Reducing Strand Flip Move

For simplicity, the following algorithm description focuses solely on strands that pass exclusively over crossings and is analogous to the case of strands passing exclusively under crossings.

For every pair of nodes in a diagram (vertices and crossings):

1. If paths passing exclusively over crossings exist, find the one containing the largest number of crossings.

2. Compute all possible paths between the nodes and find the one that contains the smallest number of crossings.

3. If the number of crossings in the minimal-crossing path (2) is strictly smaller than in the maximal over-type path (1), a diagram reducing strand flip move is possible.

Finding an Over-Type Path

Finding an 'over'-type path between two nodes is performed by following each strand incident to the starting node and tracing it through the diagram. The process terminates either unsuccessfully, if a vertex or an 'under' crossing is encountered, or successfully if the target node is reached. In the latter case, the traced path is the desired 'over'-type path.

Finding a Minimal-Crossing Path

Computing an alternative placement for a strand, together with its cost, which represents the number of crossings that would be created by inserting a strand along this path, can be obtained by:

1. Following an arc until a crossing or a vertex is reached

2. At a vertex, the two remaining arcs must be considered. These correspond to continuing the traversal with either a left or a right turn.

3. At a crossing, three arcs may be followed: one to the left, one to the right and one passing straight through the crossing.

The side of the path along which the strand is inserted determines whether a crossing is created. Taking this into account, the following cost model is defined:

- Turns at vertices or crossings have cost 1 if the side changes and cost 0 otherwise.

- Passing straight through a crossing preserves the side and has cost 1.

The minimal-crossing path is then obtained by selecting the path with the minimal total cost according to this cost model.

5 Results

All merged through pipeline2:
33 yamada groups

Table 1: Bonded knots and their Yamada polynomials

Name	Chirality	Yamada
A		$-A^{16}+A^{14}-A^{13}+2A^{12}+A^{11}-A^{10}+A^{9}-3A^{8}-3A^{6}-A^{5}-A^{3}+A-1$
B		$-A^{11}-A^{10}-A^{9}-4A^{8}+A^{7}-5A^{6}+3A^{5}-3A^{4}+3A^{3}+A+1$
C		$-A^{16}-A^{13}+A^{12}+A^{11}+3A^{9}+3A^{7}+A^{5}+A^{4}-A^{3}-1$
D		$A^{16}-A^{15}+2A^{14}+A^{13}+2A^{11}-2A^{10}+A^{9}-2A^{8}+A^{7}+A^{6}+A^{5}+2A^{4}+2A^{3}+A^{2}+A+1$
E		$A^{15}+2A^{12}+A^{11}+A^{10}+3A^{9}+3A^{7}+2A^{5}+A^{4}-A^{3}+A^{2}-A-1$
F		$A^{16}+2A^{14}+A^{13}+2A^{11}-A^{10}+2A^{9}-A^{8}+A^{7}+A^{4}+A^{3}+A^{2}+A+1$
G		$-A^{8}+A^{7}-3A^{6}+2A^{5}-4A^{4}+2A^{3}-3A^{2}+A-1$
H		$-2A^{15}-2A^{13}-2A^{12}-2A^{10}+A^{9}-A^{8}+A^{7}+A^{6}+2A^{4}+A^{3}+A^{2}+A+1$
I		$-A^{16}-A^{14}-2A^{13}-A^{11}-2A^{10}-3A^{8}-2A^{6}+A^{4}-A^{3}+A^{2}-1$
J		$-A^{16}+A^{15}+A^{14}-A^{13}+2A^{12}+A^{11}+3A^{9}+3A^{7}+A^{5}+2A^{4}-A^{3}+A^{2}+A-1$
K		$-A^{13}+A^{12}+A^{9}-2A^{8}-3A^{6}-A^{5}-2A^{4}-2A^{3}-A^{2}-A-1$
L		$A^{15}+A^{14}+2A^{12}+A^{9}-A^{8}+A^{7}-A^{6}-2A^{3}-A-1$
M		$A^{13}+A^{12}+A^{11}+2A^{10}+2A^{9}+A^{8}+3A^{7}+2A^{5}-A^{4}-A+1$
N		$A^{12}-A^{11}+2A^{10}+2A^{9}+A^{8}+6A^{7}+6A^{5}+3A^{3}+2A^{2}+2$
O		$A^{15}+A^{14}-A^{13}+A^{12}-A^{11}-2A^{10}-3A^{8}-3A^{6}-A^{5}-A^{4}-2A^{3}-1$
P		$-A^{16}+A^{14}-A^{13}+A^{12}-2A^{10}-3A^{8}-2A^{6}-A^{5}-2A^{3}-A^{2}-1$
R		$-A^{15}-A^{14}-A^{13}-A^{12}-2A^{11}-A^{9}-A^{8}+A^{7}-A^{6}+2A^{5}+2A^{3}+2A^{2}+2$
S		$-A^{11}-A^{10}-3A^{8}+3A^{7}-3A^{6}+5A^{5}-A^{4}+4A^{3}+A^{2}+A+1$
T		$-A^{16}+A^{15}-A^{13}-A^{11}-3A^{10}-3A^{8}+A^{7}-A^{6}+A^{5}+2A^{4}-A^{3}+A^{2}-1$
U		$-2A^{16}-2A^{13}+A^{12}+A^{11}-A^{10}+2A^{9}-2A^{8}+A^{7}-2A^{6}-A^{5}+A^{4}-2A^{3}+A-1$
W		$A^{13}-A^{9}-3A^{8}-A^{7}-4A^{6}-2A^{4}+A^{3}+A^{2}+2$
X		$A^{12}-A^{11}-4A^{8}+A^{7}-6A^{6}+A^{5}-4A^{4}-A+1$
Y		$A^{16}+A^{15}+A^{14}+A^{13}+A^{12}+A^{9}-A^{8}+2A^{7}-A^{6}+2A^{5}+A^{3}+2A^{2}+1$
Z		$A^{16}-2A^{11}-A^{8}+2A^{7}-A^{6}+2A^{5}-A^{4}+A^{2}-2A+1$
Ab		$-2A^{13}-A^{11}-A^{10}+2A^{9}+4A^{7}+A^{6}+3A^{5}+A^{4}-1$
Bb		$-A^{13}-A^{11}-A^{10}+A^{9}-A^{8}+2A^{7}-A^{6}+A^{5}-A^{4}-A^{3}-A^{2}-A-1$
Cc		$A^{13}+A^{12}+A^{11}+A^{10}+A^{9}-A^{8}+A^{7}-2A^{6}+A^{5}-A^{4}+A^{3}+A^{2}+1$
Dd		$-A^{16}+A^{15}-2A^{13}+A^{12}-A^{11}-2A^{10}+A^{9}-2A^{8}+2A^{7}-A^{6}+A^{5}+A^{4}-2A^{3}-2$
Ee		$A^{16}+A^{15}+A^{14}+2A^{13}+2A^{12}+A^{11}+A^{10}+A^{9}-2A^{8}+A^{7}-2A^{6}+2A^{5}+A^{3}+2A^{2}-A+1$
Ff		$A^{16}-2A^{15}+A^{14}-A^{12}+2A^{11}-A^{10}+2A^{9}-A^{8}-2A^{5}+1$
Gg		$-A^{10}+2A^{9}-7A^{8}+5A^{7}-14A^{6}+6A^{5}-14A^{4}+5A^{3}-7A^{2}+2A-1$
Hh		$-A^{14}-A^{12}-A^{11}-A^{10}-2A^{8}+A^{7}-2A^{6}+A^{5}-A^{4}+A^{2}-A+1$
Ii		$2A^{12}+2A^{10}+3A^{9}+6A^{7}+6A^{5}+A^{4}+2A^{3}+2A^{2}-A+1$

Table 2: PD codes of bonded knots

Name	PD Code
A	V[0,1,2],V[0,3,4],V[2,5,6],V[3,6,7],X[1,4,8,9],X[10,11,12,8],X[7,13,11,10],X[9,12,13,5]
B	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],V[10,9,11],V[4,12,5],X[11,8,6,12],X[3,2,7,10]
C	V[0,1,2],V[0,3,4],V[3,5,6],V[6,7,8],X[9,10,11,2],X[8,12,10,13],X[12,7,5,11],X[1,4,13,9]
D	V[0,1,2],V[3,0,4],V[5,6,1],V[3,7,5],X[8,9,10,2],X[11,12,9,8],X[4,10,13,7],X[12,11,6,13]
E	V[0,1,2],V[0,3,4],V[2,5,6],V[3,7,8],X[4,9,10,1],X[11,12,9,8],X[12,13,5,10],X[7,6,13,11]
F	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[3,2,8,9],X[10,11,9,8],X[12,13,11,10],X[13,12,6,7]
G	V[0,1,2],V[1,3,4],V[0,5,6],V[4,7,8],V[2,8,5],V[6,7,3]
H	V[0,1,2],V[0,3,4],V[5,6,7],V[3,5,8],X[2,9,10,6],X[11,12,7,10],X[12,11,13,8],X[4,13,9,1]
I	V[0,1,2],V[0,3,4],V[5,6,7],V[8,7,9],X[10,5,1,4],X[8,11,12,2],X[3,12,11,13],X[13,9,6,10]
J	V[0,1,2],V[3,4,5],V[6,0,7],V[6,3,8],X[9,10,11,2],X[5,9,12,13],X[13,12,1,8],X[7,11,10,4]
K	V[0,1,2],V[1,3,4],V[5,6,7],V[8,9,6],X[0,8,5,3],X[10,11,9,2],X[4,7,11,10]
L	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],X[6,9,10,11],X[12,10,8,13],X[13,7,5,4],X[11,12,3,2]
M	V[0,1,2],V[0,3,4],V[3,5,6],V[6,7,8],X[4,8,9,1],X[2,10,11,5],X[7,11,10,9]
N	V[0,1,2],V[3,4,5],V[6,7,4],V[2,8,9],V[0,10,3],V[10,9,6],X[1,5,11,12],X[7,8,12,11]
O	V[0,1,2],V[0,3,4],V[3,5,6],V[7,8,9],X[2,10,11,5],X[9,12,10,1],X[13,11,12,8],X[6,13,7,4]
P	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[8,9,10,11],X[12,13,9,6],X[7,10,13,12],X[2,8,11,3]
R	V[0,1,2],V[3,4,1],V[5,6,7],V[5,8,9],X[6,2,4,10],X[10,11,12,7],X[13,3,0,9],X[8,12,11,13]
S	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],V[10,11,8],V[4,7,5],X[9,11,12,6],X[2,12,10,3]
T	V[0,1,2],V[3,4,5],V[5,6,1],V[3,0,7],X[8,9,10,11],X[7,12,11,4],X[6,10,9,13],X[13,8,12,2]
U	V[0,1,2],V[3,4,5],V[0,3,6],V[6,7,8],X[2,9,10,4],X[11,10,9,12],X[8,13,12,1],X[5,11,13,7]
W	V[0,1,2],V[3,4,5],V[6,3,0],V[6,7,8],X[9,10,1,5],X[7,2,10,11],X[11,9,4,8]
X	V[0,1,2],V[3,4,5],V[6,7,8],V[9,0,6],V[10,5,1],V[9,11,10],X[4,12,7,2],X[11,8,12,3]
Y	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[2,8,9,3],X[10,11,12,13],X[8,12,11,9],X[7,10,13,6]
Z	V[0,1,2],V[3,4,5],V[6,7,8],V[6,3,0],X[9,10,1,5],X[11,9,4,8],X[10,12,13,2],X[7,13,12,11]
Ab	V[0,1,2],V[3,4,5],V[6,7,8],V[0,6,3],X[2,9,10,7],X[5,11,9,1],X[8,10,11,4]
Bb	V[0,1,2],V[0,3,4],V[3,5,6],V[5,7,8],X[8,9,10,6],X[4,10,11,1],X[2,11,9,7]
Cc	V[0,1,2],V[0,3,4],V[5,6,7],V[8,7,9],X[9,10,3,2],X[1,11,5,8],X[10,6,11,4]
Dd	V[0,1,2],V[3,4,5],V[0,3,6],V[6,7,8],X[9,10,4,2],X[11,12,10,9],X[1,8,13,11],X[12,13,7,5]
Ee	V[0,1,2],V[3,0,4],V[5,6,1],V[3,7,5],X[2,8,9,10],X[8,11,12,9],X[7,4,10,13],X[13,12,11,6]
Ff	V[0,1,2],V[3,4,5],V[6,7,8],V[6,3,0],X[5,9,10,1],X[8,11,9,4],X[2,10,12,13],X[13,12,11,7]
Gg	V[0,1,2],V[0,3,4],V[1,5,6],V[2,7,8],V[3,8,9],V[4,10,5],V[6,11,7],V[9,11,10]
Hh	V[0,1,2],V[3,4,5],V[0,6,7],V[7,8,9],X[6,2,3,10],X[10,11,12,8],X[9,13,4,1],X[13,12,11,5]
Ii	V[0,1,2],V[3,4,5],V[6,7,4],V[8,2,9],V[10,3,0],V[10,8,6],X[5,11,12,1],X[9,12,11,7]

Some diagrams were reduced to less than 8 singularities

Only 8

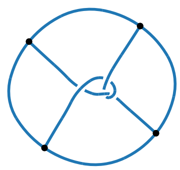
Table 3: Only 8: Bonded knots and their Yamada polynomials

Name	Chirality	Yamada
A		$-A^{16}+A^{14}-A^{13}+2A^{12}+A^{11}-A^{10}+A^{9}-3A^{8}-3A^{6}-A^{5}-A^{3}+A-1$
B		$-A^{11}-A^{10}-A^{9}-4A^{8}+A^{7}-5A^{6}+3A^{5}-3A^{4}+3A^{3}+A+1$
C		$-A^{16}-A^{13}+A^{12}+A^{11}+3A^{9}+3A^{7}+A^{5}+A^{4}-A^{3}-1$
D		$A^{16}-A^{15}+2A^{14}+A^{13}+2A^{11}-2A^{10}+A^{9}-2A^{8}+A^{7}+A^{6}+A^{5}+2A^{4}+2A^{3}+A^{2}+A+1$
E		$A^{15}+2A^{12}+A^{11}+A^{10}+3A^{9}+3A^{7}+2A^{5}+A^{4}-A^{3}+A^{2}-A-1$
F		$A^{16}+2A^{14}+A^{13}+2A^{11}-A^{10}+2A^{9}-A^{8}+A^{7}+A^{4}+A^{3}+A^{2}+A+1$
H		$-2A^{15}-2A^{13}-2A^{12}-2A^{10}+A^{9}-A^{8}+A^{7}+A^{6}+2A^{4}+A^{3}+A^{2}+A+1$
I		$-A^{16}-A^{14}-2A^{13}-A^{11}-2A^{10}-3A^{8}-2A^{6}+A^{4}-A^{3}+A^{2}-1$
J		$-A^{16}+A^{15}+A^{14}-A^{13}+2A^{12}+A^{11}+3A^{9}+3A^{7}+A^{5}+2A^{4}-A^{3}+A^{2}+A-1$
L		$A^{15}+A^{14}+2A^{12}+A^{9}-A^{8}+A^{7}-A^{6}-2A^{3}-A-1$
N		$A^{12}-A^{11}+2A^{10}+2A^{9}+A^{8}+6A^{7}+6A^{5}+3A^{3}+2A^{2}+2$
O		$A^{15}+A^{14}-A^{13}+A^{12}-A^{11}-2A^{10}-3A^{8}-3A^{6}-A^{5}-A^{4}-2A^{3}-1$
P		$-A^{16}+A^{14}-A^{13}+A^{12}-2A^{10}-3A^{8}-2A^{6}-A^{5}-2A^{3}-A^{2}-1$
R		$-A^{15}-A^{14}-A^{13}-A^{12}-2A^{11}-A^{9}-A^{8}+A^{7}-A^{6}+2A^{5}+2A^{3}+2A^{2}+2$
S		$-A^{11}-A^{10}-3A^{8}+3A^{7}-3A^{6}+5A^{5}-A^{4}+4A^{3}+A^{2}+A+1$
T		$-A^{16}+A^{15}-A^{13}-A^{11}-3A^{10}-3A^{8}+A^{7}-A^{6}+A^{5}+2A^{4}-A^{3}+A^{2}-1$
U		$-2A^{16}-2A^{13}+A^{12}+A^{11}-A^{10}+2A^{9}-2A^{8}+A^{7}-2A^{6}-A^{5}+A^{4}-2A^{3}+A-1$
X		$A^{12}-A^{11}-4A^{8}+A^{7}-6A^{6}+A^{5}-4A^{4}-A+1$
Y		$A^{16}+A^{15}+A^{14}+A^{13}+A^{12}+A^{9}-A^{8}+2A^{7}-A^{6}+2A^{5}+A^{3}+2A^{2}+1$
Z		$A^{16}-2A^{11}-A^{8}+2A^{7}-A^{6}+2A^{5}-A^{4}+A^{2}-2A+1$
Dd		$-A^{16}+A^{15}-2A^{13}+A^{12}-A^{11}-2A^{10}+A^{9}-2A^{8}+2A^{7}-A^{6}+A^{5}+A^{4}-2A^{3}-2$
Ee		$A^{16}+A^{15}+A^{14}+2A^{13}+2A^{12}+A^{11}+A^{10}+A^{9}-2A^{8}+A^{7}-2A^{6}+2A^{5}+A^{3}+2A^{2}-A+1$
Ff		$A^{16}-2A^{15}+A^{14}-A^{12}+2A^{11}-A^{10}+2A^{9}-A^{8}-2A^{5}+1$
Gg		$-A^{10}+2A^{9}-7A^{8}+5A^{7}-14A^{6}+6A^{5}-14A^{4}+5A^{3}-7A^{2}+2A-1$
Hh		$-A^{14}-A^{12}-A^{11}-A^{10}-2A^{8}+A^{7}-2A^{6}+A^{5}-A^{4}+A^{2}-A+1$
Ii		$2A^{12}+2A^{10}+3A^{9}+6A^{7}+6A^{5}+A^{4}+2A^{3}+2A^{2}-A+1$

Table 4: Only 8: PD codes of bonded knots

Name	PD Codes
A	V[0,1,2],V[0,3,4],V[2,5,6],V[3,6,7],X[1,4,8,9],X[10,11,12,8],X[7,13,11,10],X[9,12,13,5]
B	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],V[10,9,11],V[4,12,5],X[11,8,6,12],X[3,2,7,10]
C	V[0,1,2],V[0,3,4],V[3,5,6],V[6,7,8],X[9,10,11,2],X[8,12,10,13],X[12,7,5,11],X[1,4,13,9]
D	V[0,1,2],V[3,0,4],V[5,6,1],V[3,7,5],X[8,9,10,2],X[11,12,9,8],X[4,10,13,7],X[12,11,6,13]
E	V[0,1,2],V[0,3,4],V[2,5,6],V[3,7,8],X[4,9,10,1],X[11,12,9,8],X[12,13,5,10],X[7,6,13,11]
F	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[3,2,8,9],X[10,11,9,8],X[12,13,11,10],X[13,12,6,7]
H	V[0,1,2],V[0,3,4],V[5,6,7],V[3,5,8],X[2,9,10,6],X[11,12,7,10],X[12,11,13,8],X[4,13,9,1]
I	V[0,1,2],V[0,3,4],V[5,6,7],V[8,7,9],X[10,5,1,4],X[8,11,12,2],X[3,12,11,13],X[13,9,6,10]
J	V[0,1,2],V[3,4,5],V[6,0,7],V[6,3,8],X[9,10,11,2],X[5,9,12,13],X[13,12,1,8],X[7,11,10,4]
L	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],X[6,9,10,11],X[12,10,8,13],X[13,7,5,4],X[11,12,3,2]
N	V[0,1,2],V[3,4,5],V[6,7,4],V[2,8,9],V[0,10,3],V[10,9,6],X[1,5,11,12],X[7,8,12,11]
O	V[0,1,2],V[0,3,4],V[3,5,6],V[7,8,9],X[2,10,11,5],X[9,12,10,1],X[13,11,12,8],X[6,13,7,4]
P	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[8,9,10,11],X[12,13,9,6],X[7,10,13,12],X[2,8,11,3]
R	V[0,1,2],V[3,4,1],V[5,6,7],V[5,8,9],X[6,2,4,10],X[10,11,12,7],X[13,3,0,9],X[8,12,11,13]
S	V[0,1,2],V[0,3,4],V[1,5,6],V[7,8,9],V[10,11,8],V[4,7,5],X[9,11,12,6],X[2,12,10,3]
T	V[0,1,2],V[3,4,5],V[5,6,1],V[3,0,7],X[8,9,10,11],X[7,12,11,4],X[6,10,9,13],X[13,8,12,2]
U	V[0,1,2],V[3,4,5],V[0,3,6],V[6,7,8],X[2,9,10,4],X[11,10,9,12],X[8,13,12,1],X[5,11,13,7]
X	V[0,1,2],V[3,4,5],V[6,7,8],V[9,0,6],V[10,5,1],V[9,11,10],X[4,12,7,2],X[11,8,12,3]
Y	V[0,1,2],V[0,3,4],V[1,5,6],V[4,7,5],X[2,8,9,3],X[10,11,12,13],X[8,12,11,9],X[7,10,13,6]
Z	V[0,1,2],V[3,4,5],V[6,7,8],V[6,3,0],X[9,10,1,5],X[11,9,4,8],X[10,12,13,2],X[7,13,12,11]
Dd	V[0,1,2],V[3,4,5],V[0,3,6],V[6,7,8],X[9,10,4,2],X[11,12,10,9],X[1,8,13,11],X[12,13,7,5]
Ee	V[0,1,2],V[3,0,4],V[5,6,1],V[3,7,5],X[2,8,9,10],X[8,11,12,9],X[7,4,10,13],X[13,12,11,6]
Ff	V[0,1,2],V[3,4,5],V[6,7,8],V[6,3,0],X[5,9,10,1],X[8,11,9,4],X[2,10,12,13],X[13,12,11,7]
Gg	V[0,1,2],V[0,3,4],V[1,5,6],V[2,7,8],V[3,8,9],V[4,10,5],V[6,11,7],V[9,11,10]
Hh	V[0,1,2],V[3,4,5],V[0,6,7],V[7,8,9],X[6,2,3,10],X[10,11,12,8],X[9,13,4,1],X[13,12,11,5]
Ii	V[0,1,2],V[3,4,5],V[6,7,4],V[8,2,9],V[10,3,0],V[10,8,6],X[5,11,12,1],X[9,12,11,7]

Knot Diagrams - 8 singularities



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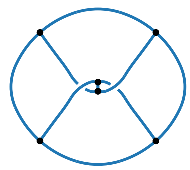

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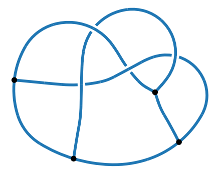

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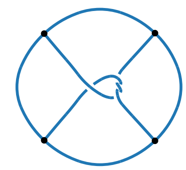

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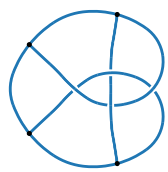

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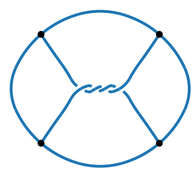

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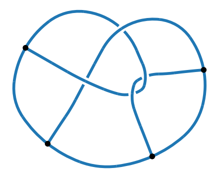

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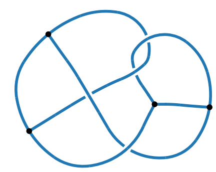

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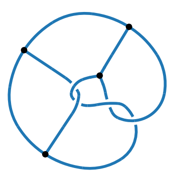

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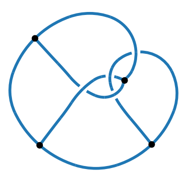

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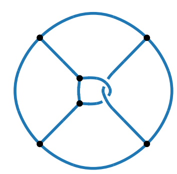

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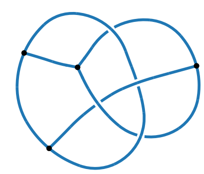

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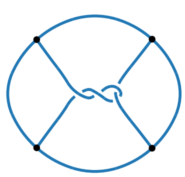

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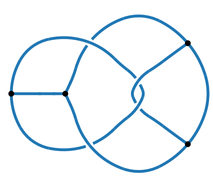

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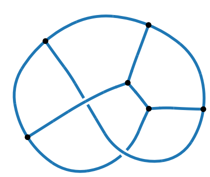

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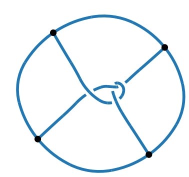

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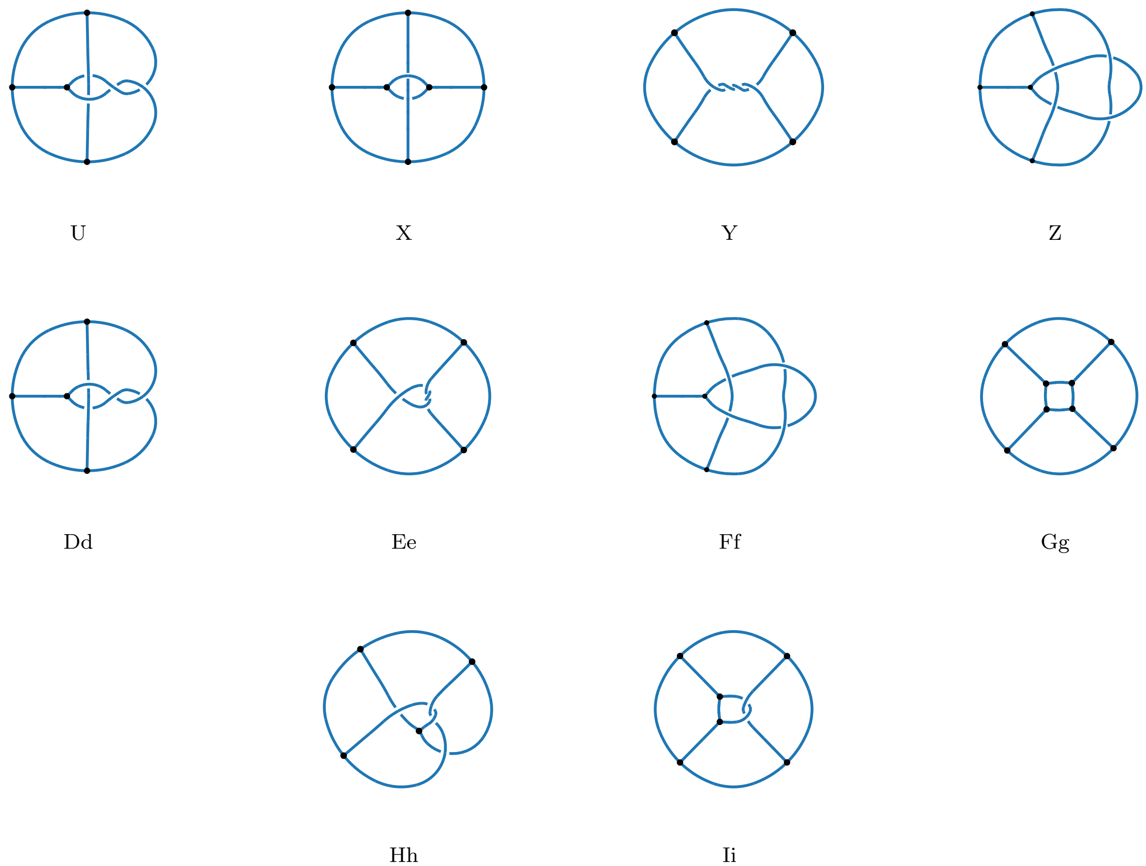

Figure 4: Bonded knots. Each entry is represented by two rows: the diagram and its label.